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For my General Education courses, I took a mixture of courses that helped me learn concepts outside of Electrical Engineering. These concepts in some shape or form helped me along the way through engineering and helped me change the way I think through engineering problems outside of technical information. For the courses I took they were Microeconomics, Russia Today, Personal Finance, and American History of Rock and Roll. Although, in many ways these courses are completely different from each other and have almost nothing to do with the engineering curriculum here at Iowa State University, I was able to find a way to take the information from these courses and turn them into something useful for my Career Path in Engineering. Due to constraints of this paper, I will focus on Microeconomics and Russia Today to show the impact of both of these courses in my engineering career.

To start out with Microeconomics, this is a course that teaches you about the basics of markets, and how they change based on the interest of the consumers in these markets. In other words, Microeconomics teaches you how supply and demand change over time and cause markets to change for companies. This helps me evaluate and formulate engineering solutions beyond the technical aspect by making me think of how my decisions can affect the markets of the business that the decisions are for. For example, let's say I work for Texas Instruments, and I am working in a team that will create a new chip that will be in the new TI calculator that will replace the infamous TI-84 calculator. When designing the chip, I have to think of factors that are outside of my technical thoughts. For example, let's say with this new calculator I make the decimal value for rounding different compared to previous calculators that TI makes, this can affect Texas Instruments Economically. Students, Professors, Researchers, Scientists, etc will all buy the calculator and depending upon the liking of these consumers, then the demand of our supply will change for better or worse. This is how my Microeconomics knowledge comes into play. If I am designing something, I need to think of the economics of it to plan out my design better so that the demand of my design is high and that the economy of my work is good and not bad. Globally, the issues we can face is if our market reaches outside the scope of the nation I work in. With the Texas Instruments example, let's say that TI started selling this calculator in other countries, well then, the company faces potential issues in a global economic standpoint that the users in countries like China, India, etc, may not like the calculator as much and then the demand is low there which can hit TI in a bad way. However, if the calculator is a success this would be great for TI because then they are

making a very high global GDP and their economics is good. Societal impacts are that the design can have an impact on the livelihood of people using the design. For example, let's say that the user of my design for a road doesn't like my design because it makes their trip to work longer, which means that they have to get up earlier to get to work, which means they get less sleep or less time with their loved ones and thus creates problems for the user that could have been prevented if the engineer came up with a better design. The flip side is like my design and then they are happy with the work which keeps my job stable.

Next, I will talk about Russia Today. Russia Today is a course that teaches students about the history of how the Russian Government has treated their people since the rise of the USSR and how the people of Russia felt about the government and their control. The overall message was that the people just wanted to live their lives and that the government did what was best for the government and not what was best for the people. Although this does not seem like a class that I can use to help me in Engineering. I think it is a class that was actually beneficial. Russia Today taught me that the clients I will have in my career at the end of the day are just people who have needs and are coming to me to fix their issues. As an engineer if I was unethical, I could use my knowledge to my advantage and thus take advantage of my clients by charging my services or not giving my all in my design. This class taught me my actions as an engineer can affect people's lives so I should always take my work with care and do the right thing for the people and not what is better for my personal greed. The impact of this course on knowledge economically is by thinking of the end users more than your personal greed, the company you work for will gain a more prestigious name because they will be known as a company people can rely on. This boosts the companies GDP and the country the company resides in GDP. The global impact is that this can be beneficial in many countries and can create better economic growth for the company and countries and would help people in other countries rather than just the origin country. The societal impact is that society will be happier overall if the engineering company treats the customers with the upmost respect in their work.

In Conclusion, general education courses have a big impact in how we as engineers make our decision and it is more than we could have ever thought. These courses help us determine other characteristics of our work that are not technical like our finances, economics, and customer policy. It helps us solve our problems with our work that we cannot solve with the knowledge we gained in our engineering courses. Thank you, Iowa State, for giving a good education to me to make me a good all-around engineer,